Exploring Math Wikipedia Page Views

2025-03-12

Research Question & Motivation

- What is the general publics interaction with Mathematics?
- We will examine this question through analysis of page view statistics for Math-related Wikipedia articles
- Outline:

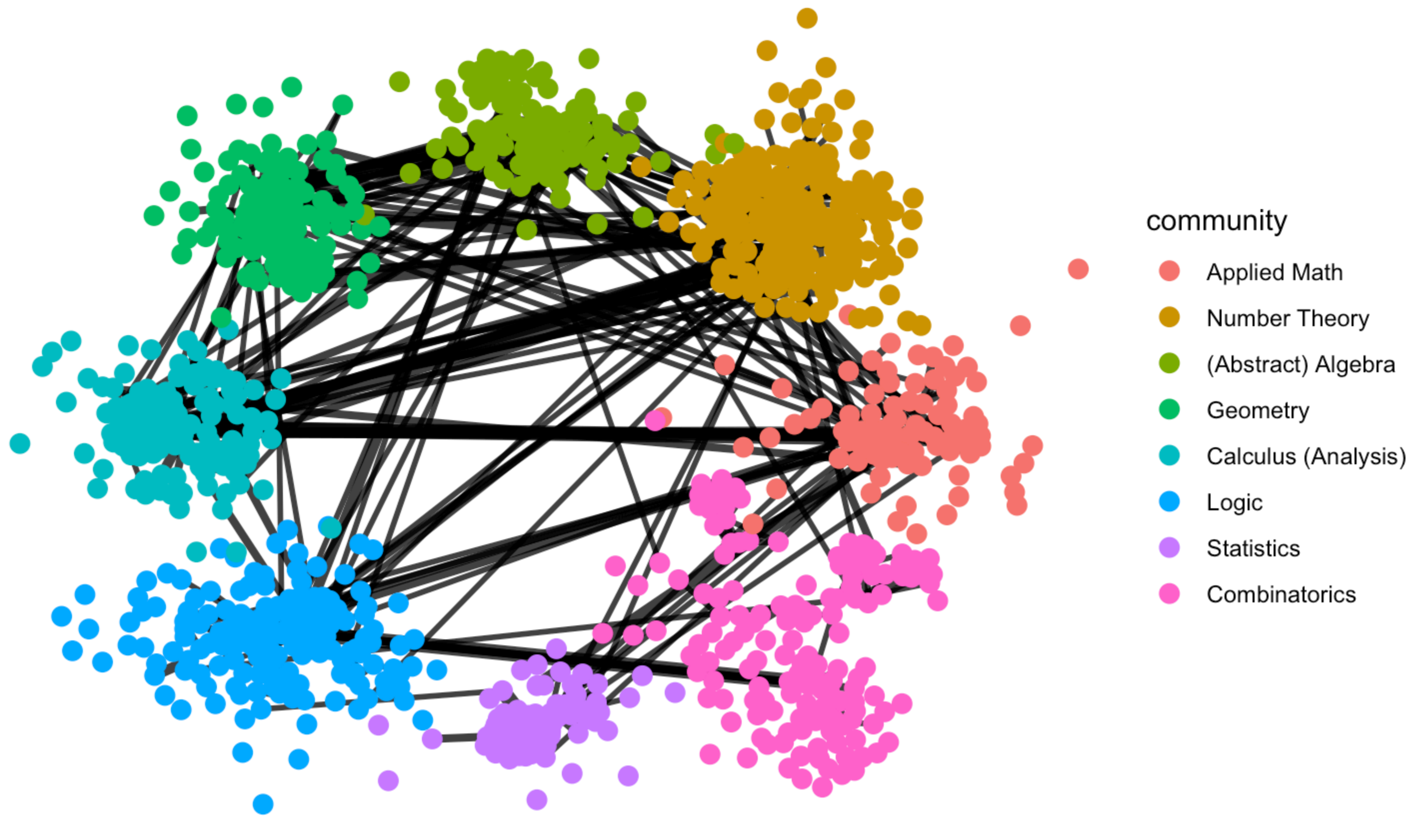
1. Overview of data set
2. Wikipedia page connection graph
3. Viewership Statistics
4. Relevant events impacting page views

Math Wikipedia Dataset

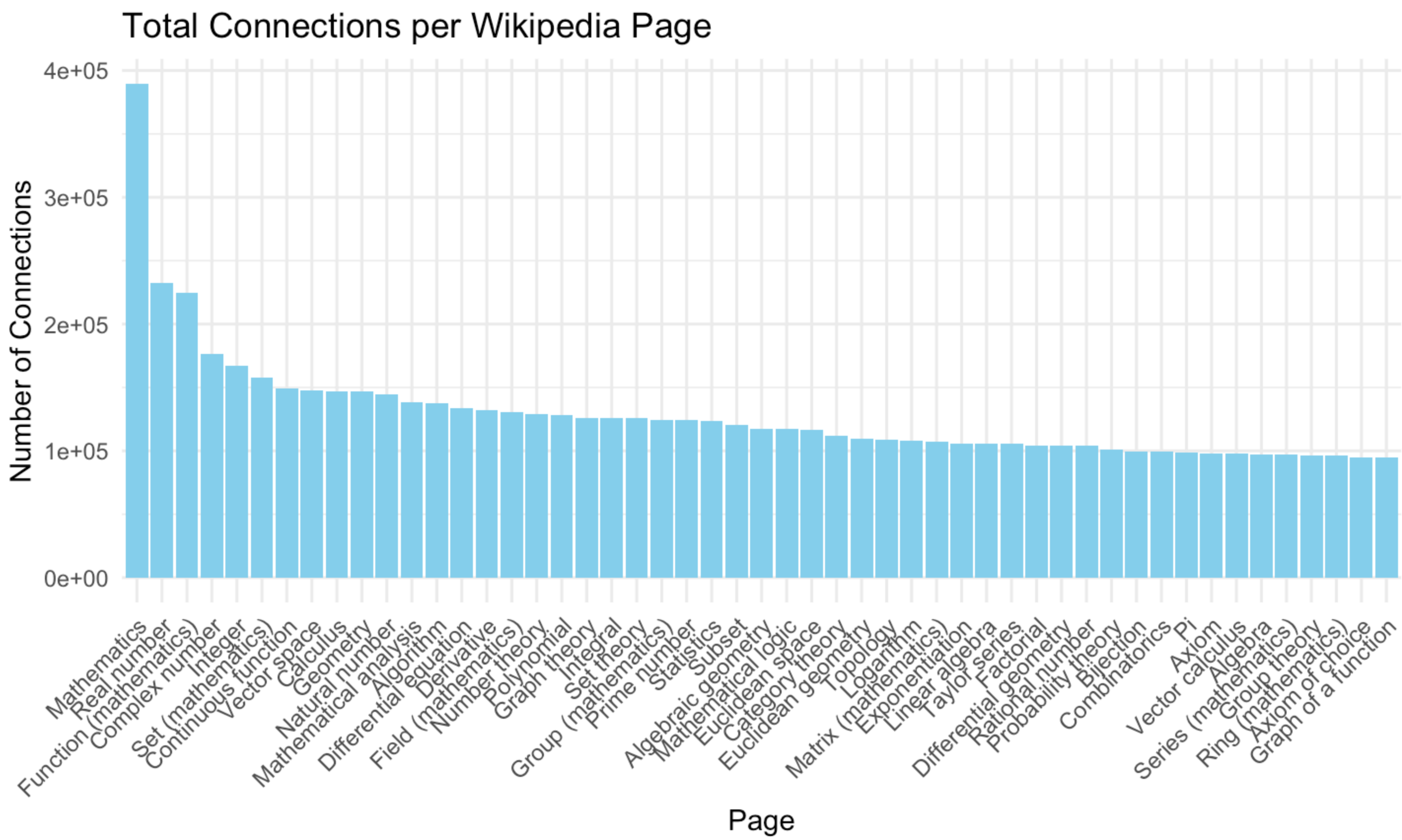
- Wikipedia Math Essentials from UCI Machine Learning Repository. (https://archive.ics.uci.edu/dataset/605/wikipedia+math+essentials)
- Daily view count data for 1068 Wikipedia pages related to Mathematics and Statistics, over period of 2 years, with one instance per page per day
- Period in question: March 16, 2019 - March 15, 2021
- Variables: Page Name, Date, View Count, Edges (number of hyperlinks with other pages)

```
##          Page ID Edges       Date Year Month Day Views
## 1 Mathematics  0   533 2019-03-16 2019     3  16  3962
## 2 Mathematics  0   533 2019-03-17 2019     3  17  3923
## 3 Mathematics  0   533 2019-03-18 2019     3  18  4617
## 4 Mathematics  0   533 2019-03-19 2019     3  19  4822
## 5 Mathematics  0   533 2019-03-20 2019     3  20  4871
## 6 Mathematics  0   533 2019-03-21 2019     3  21  4969
```

Wikipedia Page Connection Map



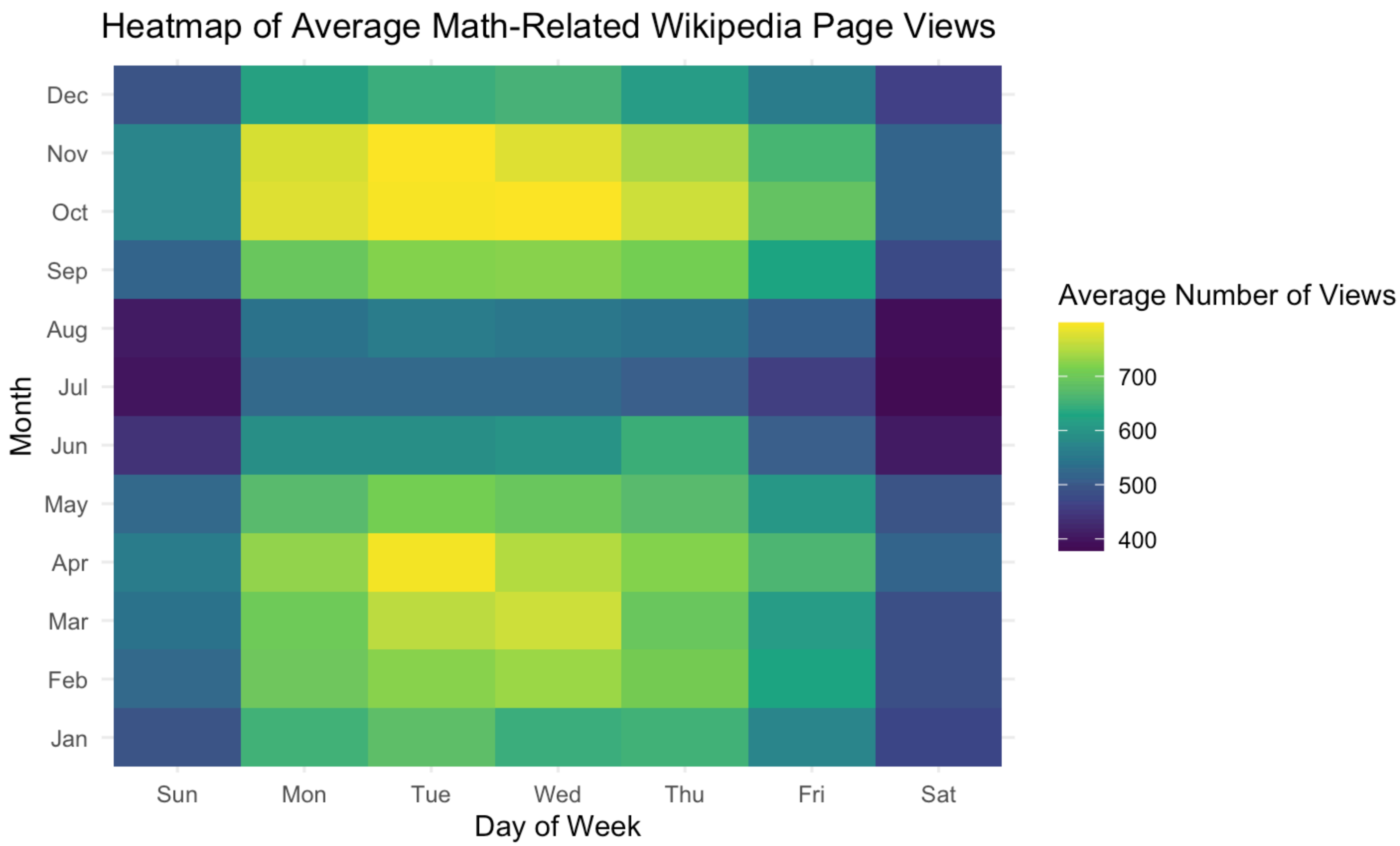
Top 50 Connected Pages



Most Viewed Pages

```
## # A tibble: 1,068 × 2
##    Page                   Views
##    <chr>                  <int>
##  1 Normal distribution  5355019
##  2 Standard deviation   5131200
##  3 Fibonacci number     4524975
##  4 Pi                   3742122
##  5 Roman numerals       3730168
##  6 Prime number         3368175
##  7 Poisson distribution 3276035
##  8 Mathematics          3192685
##  9 Golden ratio         3038084
## 10 Taylor series        2637277
## # ℹ 1,058 more rows
```

Average Number of Views per Weekday



Math Wikipedia and Media

- Question: What types of events influence public interests in Mathematics, and can we quantify this through Wikipedia page views?
- What do the sources of these sparks in interest tell us about the best way to promote Mathematics to a wider audience?
- Three cases: Math related holidays, mainstream press stories, and Youtube videos

Shiny App

https://mlewispeinad.shinyapps.io/project1/

- Allows us to filter through different Math Wikipedia pages and date ranges

Case 1: Math Holidays

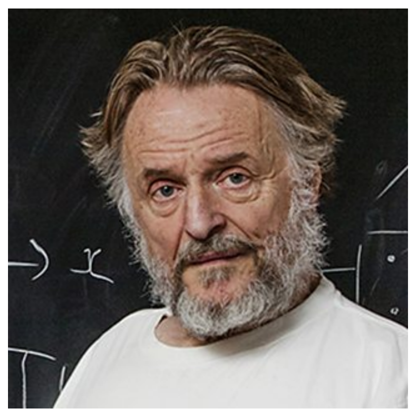
- Two notable Math holidays
- Pi Day (March 14)
- Fibonacci Day (November 23)

$$1+1=2$$
$$1+2=3$$
$$2+3=5$$
$$3+5=8$$
$$5+8=13$$
$$8+13=21$$
$$13+21=34$$
$$21+34=55$$
$$\cdots$$

The Fibonacci Sequence

Case 2: Mainstream Media Articles

- From examining archived articles from publications such as the New York Times, Math related articles are few and far between in the mainstream press.
- Common type of Math-related news article: Obituaries for famous Mathematicians
- Example: John Horton Conway (26 December 1937 – 11 April 2020), known for his contributions to both academic and recreational Mathematics, for example Coding Theory



Exponential Growth

- Early during the COVID pandemic, many people were curious about how the virus was spreading so quickly, leading to talk about the Mathematics behind ‘Exponential Growth’.

The New York Times

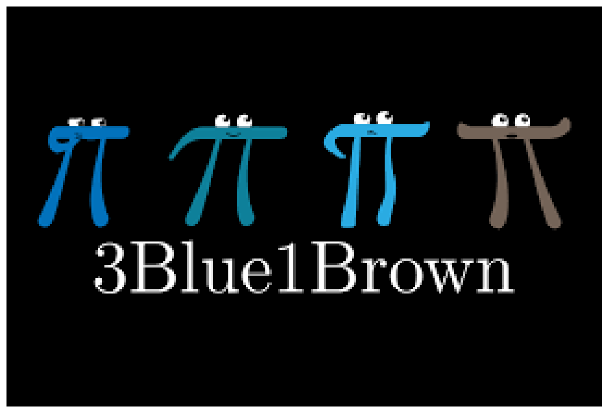
CURRENT EVENTS

The Math of Ending the Pandemic: Exponential Growth and Decay

In this lesson, students will explore how the mathematical concepts of exponential growth and exponential decay help to explain the spread and slowdown of the coronavirus.

Case 3: Youtube

- Youtube has served as a thriving community for exposing Mathematics to a general audience
- Example: 3Blue1Brown (Grant Sanderson), known for producing accessible videos on advanced topics in Math and Science with slick animations
- 7 Million subscribers as of 2025



Stanford Math Alumnus, Grant Sanderson, to Receive Communications Award from AMS | Mathematics

Conclusion

- Given the prevalence of Wikipedia in many of our everyday lives, we can learn a lot about societies relationship with Mathematics through examining trends on Wikipedia
- An important takeaway is that alternative media outlets, such Youtube, are more effective at promoting interest in Mathematics than the traditional press
- Happy early Pi Day!